ForgeMind™

Trustworthy Geospatial Answers for Mission Environments

ForgeMind™ turns a plain-language request into a validated geospatial pipeline on ForgeGIS™ — or into a named refusal when the request cannot be honored correctly. On the compiled path the language model is confined to understanding intent, and deterministic code owns what runs. It runs on one workstation beside the rest of the Forge suite, bound to loopback, and with a model server on the site's own network it sends no request to a cloud model vendor.

Refuse

rather than fabricate

on the compiled path, a request becomes a type-checked pipeline or a specific, machine-readable refusal

Local

model, if you choose

any OpenAI-compatible server on your network — Ollama is the tested path — alongside four commercial vendors

127.0.0.1

loopback by default

a per-install bearer token in the suite; unapproved origins and DNS-rebound pages refused

Why refusal matters here

In mission work a plausible-but-wrong product is worse than none. An ambiguous reference goes back to the operator as a question, not a guess. Across its June 2026 conversation validation campaign ForgeMind returned no confident-wrong answers. Where the compiler declines a request, a guarded agent loop may carry out a map action, but since 1.2.6 it never answers a question for a figure, and any figure a model writes is checked against the measured facts.

Geometry it does not re-derive

Named geometry an operator places — an objective, a route, a landing zone — is referenced by a stable handle and hydrated from the catalog when an operation needs it, never reconstructed from a bounding box. If the true geometry is not reachable, the operation defers and asks. Since 1.2.5, the scene — entities with their MIL-STD-2525 symbol codes, overlays, the scenario clock — can travel with each question.

Contained by design

ForgeGIS and ForgeData run as local child processes over stdio; the map-surface and broker adapters are local children that reach their apps over HTTP. Each sits behind a circuit breaker; a slow call fails alone and a hung worker is recycled. Every inbound message is size-capped, and a set of misconfigurations is refused at startup rather than discovered mid-mission.

New in 1.2.6 — hardened on the workstation. Without a token, a request must address ForgeMind as a loopback host, so a DNS-rebound web page cannot read it; a bind beyond loopback without a token refuses to start. The Control Center now stops ForgeMind gracefully, and a request still running is told it did not finish. Each routed question is logged to a local query-capture file by default, and one setting turns that off.

CONTINUED ON REVERSE →

WHAT PROGRAMS GET

The line-of-sight surface, by intent

Viewshed, intervisibility, slope and terrain-constrained site suitability — "where can we land outside the enemy viewshed?" — reachable in plain language and run on the GPU-accelerated engine against real elevation, with named anchors resolved against the session's ledger. Every distance, area and bearing is measured on the WGS84 ellipsoid, and a radius question no longer answers from ground outside the radius.

Answers where the operator is looking

Ask in ForgeGIS Studio™ and the answer is drawn on Studio's map; ask in Seaglass Globe™ and it lands on the globe, beside the exercise's symbols. With the compute broker installed, a folder batch of a supported raster operation goes to it as one job.

Knows what is on the shelf

New in 1.2.6, "what elevation do we have for this area?" or "is there 30 m data here?" is answered from the local ForgeData catalog: per kind, how many datasets, the finest resolution, sources and dates. A yes/no question leads Yes, No or Partly, and a stated resolution is checked, so 90 m data is never a yes to "30 m".

Nothing stated that was not measured

Since 1.2.6 any length, area, angle or percentage a model writes into a reply must match a measured fact, or the sentence is replaced or cut, and a claim that something is on the map needs a map call that succeeded. An answer over an area placed from the model's memory, or at a place ForgeMind cannot check, says so.

A model that stays on site

Run an open-weight model on your own hardware behind an OpenAI-compatible server and point ForgeMind at it; the Seaglass Control Center™ can check that the model actually answers before it saves. Expect a local model to be slower than a hosted one; in the suite it runs every stage on the one model.

> **What the figures rest on.** The zero-confident-wrong result is the June 2026 conversation campaign, run before 1.2.5's provider options existed; the published 102-operation marathon session is a scripted regression run on 1.2.5 (September 2026) in which the test harness supplies each turn's intent. OpenAI, Gemini and Grok on the compiled path were exercised against a test stub rather than live keys. The Technical Brief sets out exactly what each figure shows.

Request the Technical Brief and an evaluation build

The Technical Brief covers the architecture, the refusal model and where it stops, providers, deployment and security. ForgeMind is part of the Forge suite, installed by the Seaglass Control Center. The Forge suite is commercially licensed; evaluation builds are available to qualified programs.

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